

Question1: Write the program to compute mean, median, mode, standard deviation, variance, correlation, covariance of given data table format 10X3 matrix where C1 and C2 has integer data type and C3 has character type.

Program

```
import pandas as pd
import numpy as np
import math

marks = [98,56, "P" ,78,43, "P", 56,23, "F", 67,87, "P", 99,54, "P", 78,27, "F", 77,45, "P", 56,57,
"P", 89,34, "F" ,44,43, "P"]

data = np.array(marks).reshape(10,3)

results_mapping = {
    "P": 1,
    "F": 0
}

df = pd.DataFrame(data, columns=["sub1", "sub2", "result"])
df['sub1'] = pd.to_numeric(df['sub1'])
df['sub2'] = pd.to_numeric(df['sub2'])

df['result'] = df['result'].map(results_mapping)
df['result'] = pd.to_numeric(df['result'])

print("Mean")
print(df[['sub1', 'sub2', 'result']].mean())

print("\nMedian:")
print(df[['sub1', 'sub2', 'result']].median())

print("\nMode:")
print(df[['sub1', 'sub2', 'result']].mode())

print("\nStandard Deviation:")
print(df[['sub1', 'sub2', 'result']].std())

print("\nVariance:")
print(df[['sub1', 'sub2', 'result']].var())

print("\nCorrelation between sub1 and sub2:")
print(df['sub1'].corr(df['sub2']))
```

```

print("\nCovariance Matrix:")
print(df[['sub1', 'sub2', 'result']].cov())

print("\nResult Distribution:")
print(df['result'].value_counts())

print("\nOverall Mean")
print(df.mean().mean())

print("\nOverall Median")
print(df.median().median())

print("\nOverall Mode")
print(df.mode().mode())

print("\nOverall Standard Deviation")
print(df.std().std())

print("\nOverall Variance")
print(df.var().var())

```

Output:

```

Mean
sub1    74.2
sub2    46.9
result    0.7
dtype: float64

Median:
sub1    77.5
sub2    44.0
result    1.0
dtype: float64

Mode:
  sub1  sub2  result
0   56  43.0    1.0
1   78   NaN    NaN

Standard Deviation:
sub1    18.449932
sub2    18.290556
result    0.483046
dtype: float64

Variance:

```

```
sub1    340.400000
sub2    334.544444
result   0.233333
dtype: float64
```

Correlation between sub1 and sub2:
0.061307764484815494

Covariance Matrix:

```
      sub1    sub2  result
sub1  340.400000  20.688889 -0.044444
sub2  20.688889  334.544444  6.300000
result -0.044444   6.300000  0.233333
```

Result Distribution:

```
result
```

```
1    7
```

```
0    3
```

Name: count, dtype: int64

Overall Mean

40.6

Overall Median

44.0

Overall Mode

```
      sub1  sub2  result
```

```
0    56  43.0    1.0
```

```
1    78   NaN    NaN
```

Overall Standard Deviation

10.327486205085725

Overall Variance

37918.59460905351

Program2: Draw a BarPlot for the following data of company and its profit. Also draw BoxPlot also.

Program:

```
import matplotlib.pyplot as plt
import numpy as np

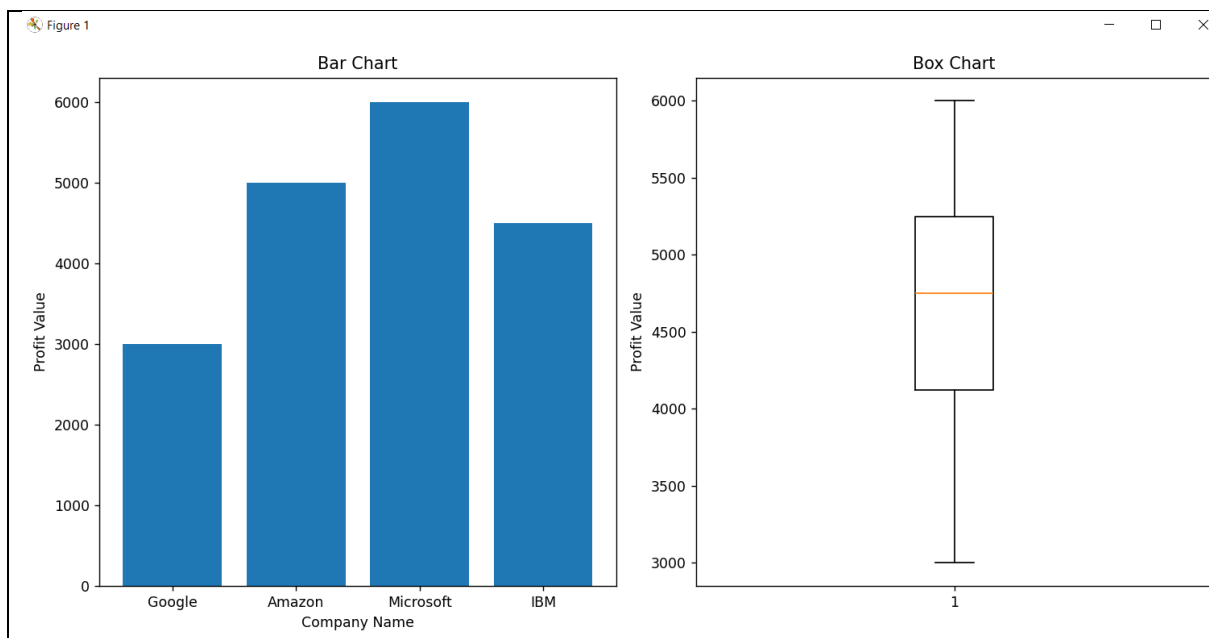
companyname = ["Google", "Amazon", "Microsoft", "IBM"]
profit = [3000, 5000, 6000, 4500]
plt.figure(figsize=(12,6))
```

```
plt.subplot(1,2,1)
plt.bar(companyname, profit)
plt.title("Bar Chart")
plt.xlabel("Company Name")
plt.ylabel("Profit Value")

plt.subplot(1,2,2)
plt.boxplot([profit], label=["Companies"])
plt.title("Box Chart")
plt.ylabel("Profit Value")

plt.tight_layout()
plt.show()
```

Output:



Program 3: Draw a line plot and scatter plot for the following data

Semester -> [1,2,3,4,5,6,7,8]

Grades -> [A+, A, A, D, B+, B, A, A]

Program:

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    "Semester" : [1,2,3,4,5,6,7,8],
    "Grades" : ["A+", "D", "A", "B+", "A+", "A", "D", "B+"]
}
```

```

df = pd.DataFrame(data)

print(df)

grade_mapping = {
    'A+' : 10,
    'A' : 9,
    'B+' : 8,
    'B' : 7,
    'D' : 4
}

#Converting grades to numeric values
df['Numeric_Grades'] = df['Grades'].map(grade_mapping)

plt.figure(figsize=(12,5))

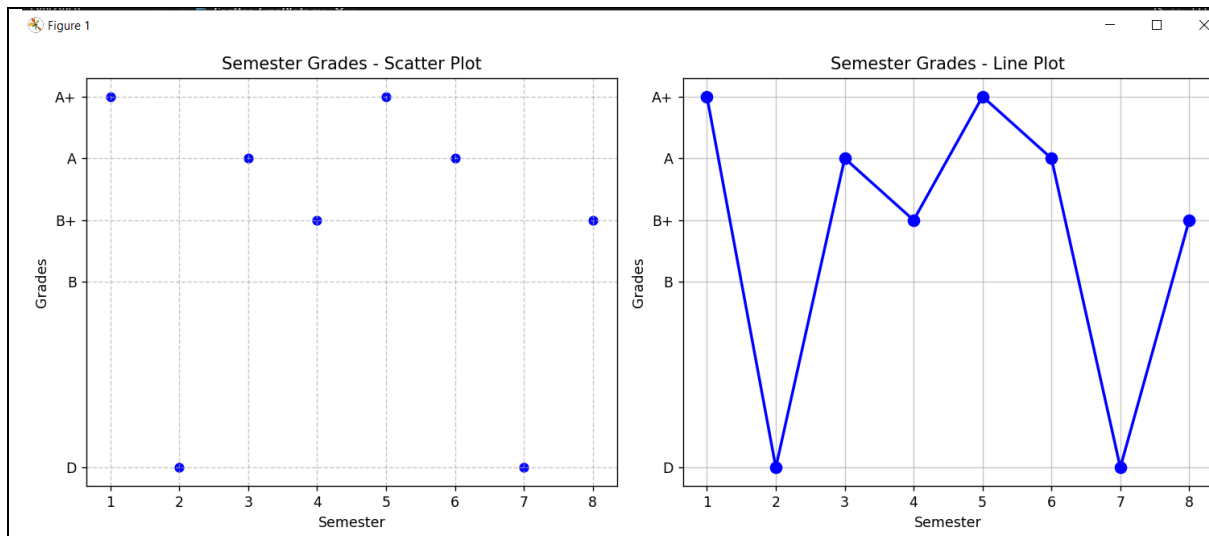
plt.subplot(1,2,1)
plt.scatter(df['Semester'], df['Numeric_Grades'], color='blue', marker='o')
plt.title("Semester Grades - Scatter Plot")
plt.xlabel("Semester")
plt.ylabel("Grades")
plt.grid(True, linestyle="--", alpha= 0.7)
plt.yticks(list(grade_mapping.values()), list(grade_mapping.keys()))

plt.subplot(1,2,2)
plt.plot(df['Semester'], df['Numeric_Grades'], 'r-o',color='blue', linewidth=2, markersize = 8)
plt.title("Semester Grades - Line Plot")
plt.xlabel("Semester")
plt.ylabel("Grades")
plt.yticks(list(grade_mapping.values()), list(grade_mapping.keys()))
plt.grid(True, linestyle="--", alpha= 0.7)
plt.tight_layout()

plt.show()

```

Output:



Program4: Clean the dataset with replacing all the column values which contains ? to NAN, create dataset 5X3 and update our csv file

Program

```
import numpy as np
import pandas as pd

pd.set_option('future.no_silent_downcasting', True)

data = {
    'Age': [25, '?', 30, 35, '?', 28],
    'Income': ['?', 45000, 55000, '?', 65000, 48000],
    'Education': ['Bachelors', '?', 'Masters', 'PhD', 'Bachelors', '?']
}

df = pd.DataFrame(data)

df = df.replace('?', np.nan)

print("\nDataFrame after replacing '?' with NaN:")
print(df)

print("\nData Info after cleaning:")
print(df.info())

print("\nNumber of NaN values in each column:")
print(df.isna().sum())

#Save the cleaned dataset
df.to_csv("cleaned_dataset.csv", index=False)
```

Output:

Age	Income	Education
25	45000	Bachelors
30	55000	Masters
35	65000	PhD
	48000	Bachelors
28		